Capping the Capacity Market

A Supply Function Equilibrium Analysis of Price Controls in PJM

April 21, 2026

PJM at a Glance

- Regional transmission organization covering 13 states plus DC, about 65 million customers and 180 GW of capacity
- Operates three wholesale markets
  - Energy pays for electricity produced
  - Ancillary services pay for reserves and grid support
  - Capacity pays for availability three years ahead
- This paper studies the capacity market (annual BRA under RPM)

The $21 Billion Question

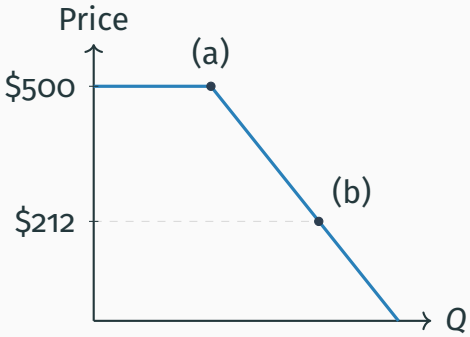
- In December 2024, Pennsylvania filed a FERC complaint after the 2025/26 auction cleared at $269.92/MW-day, an 800% jump
- In January 2025, the Shapiro-PJM settlement capped the 2026/27 and 2027/28 auctions at $325/MW-day with a $175/MW-day floor
- Settlement claims over $21 billion in two-year consumer savings

Research Question

What would PJM's 2026/27 and 2027/28 auctions have cleared at **absent the settlement**?

And what is the actual revenue transfer it produces?

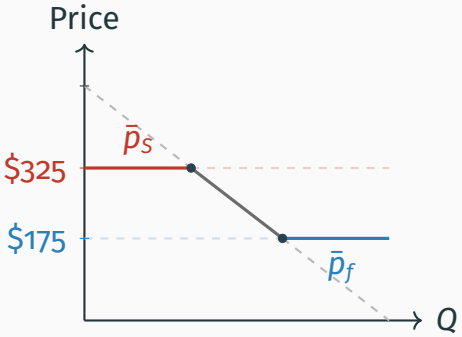
The VRR Curve Slopes Only in Scarcity



Source: stylized from PJM 2026/27 VRR parameters

- Flat above (a) at the administrative cap
- Slopes from (a) through (b) at Net CONE in the scarcity region
- No pre-settlement floor: curve continues to fall

The Settlement Bounds the Clearing Price



Source: Shapiro–PJM settlement (Jan 2025) on the stylized VRR curve

- Cap at $325 truncates from above
- Floor at $175 supports from below
- Applies only to 2026/27 and 2027/28, RTO and LDAs

Model: Supply Function Equilibrium

K symmetric sellers bid supply functions against the VRR demand, in the tradition of Klemperer–Meyer (1989)

$$s'(p) \;=\; \frac{D'(p) \;+\; s(p)/(p-c)}{K-1}$$

- Holmberg (2008) boundary $s(\bar{p}) = \bar{q}$ selects the equilibrium
- Competitive fringe offers a block Q_f at cost c
- p^* solves $K \cdot s(p^*) + Q_f = D(p^*)$

The At-Cap Regime Is Mechanical

- When $K \leq 3$, the interior ODE cannot clear demand below $\bar{p}$
- The market then clears at $\bar{p}$ *by construction* via the boundary condition
- The \$329 and \$333 figures come from PJM's VRR Point (a), not from an independent price forecast

The substantive content is the *regime* (at-cap vs. interior) and its dependence on K

Calibration

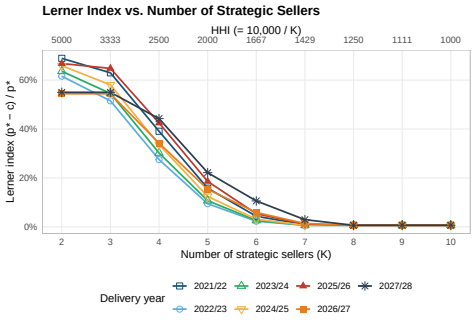
- Number of strategic sellers set to $K = 3$
  - All seven BRA years show $\text{RSI}_3 < 1$ at the RTO level
- Marginal cost set to $c = \$150/\text{MW-day}$
  - Combined-cycle sector ACR from IMM (2025); this paper's choice, not the IMM's own recommended benchmark
- VRR curve, demand, and CETL drawn from PJM planning filings

Baseline SFE: New Design Years Clear at the Cap

Year	Design	p^*	Actual	Margin	Outcome
2021/22	Old	\$405	\$140	+6.8%	Interior
2022/23	Old	\$310	\$50	+9.2%	Interior
2023/24	Old	\$330	\$34	+9.9%	Interior
2024/25	Old	\$358	\$29	+11.7%	Interior
2025/26	Old	\$426	\$270	+1.6%	Interior
2026/27	**New**	**\$329**	**\$329**	−0.2%	**At cap**
2027/28	**New**	**\$333**	**\$333**	−4.7%	**At cap**

Source: PJM BRA reports; p^* from SFE model at K=3, c=\$150. Margin = (cleared − reliability req.)/reliability req.; new-design years cleared short

Markups Collapse Between $K = 3$ and $K = 4$



Source: SFE Lerner index $L = (p^* - c)/p^*$ from numerical ODE solution at $K \in \{2, \ldots, 10\}$; 2026/27 PJM parameters, baseline calibration

Same Arithmetic, Different Counterfactual

Counterfactual	2026/27	2027/28	Two-year total
VRR Point (a) [Shapiro]	$500	$500	**$17.16B**
SFE equilibrium [this paper]	$329	$333	**$0.61B**
Settlement cap	$325	$325	—

Source: Savings = (counterfactual − cap) × MW × 365, RTO level; constrained LDAs add more. *Caveat:* both years cleared short of reliability req., so at-cap is overdetermined and magnitude is Point (a)-sensitive

The $21B figure depends on the counterfactual chosen

Limitations

- New-design years cleared short of reliability req.: at-cap is overdetermined (strategic mechanism vs. physical scarcity)
- TPS mitigation not endogenous, may be the true binding constraint
- Firms assumed symmetric (cost heterogeneity bracketed)
- Holmberg boundary used as selection rule, not derived
- Static model - no retirement or entry response

What the Settlement Does, and What It Does Not

- Cap $325 sits 53% above Net CONE ($212) - entry remains viable, so the concern is informational not cost recovery
- Structural channels (transmission, steeper VRR, entry) reduce markups without truncating the signal
- Temporary suppression is not a substitute for structural reform

Thank you